# Arm actions at 3.5 Hz vs 1 Hz: what the references demand and what the policies do

Companion to `action_scale.pdf`. pal_mjlab `omm/action-scale-graphs`, October 2026.

**Notation.** Setpoint $q_{sp} = q_0 + \alpha a$ with $\alpha = 0.25\,\tau_{\max}/K_p$ (rad per action unit). The action-rate penalty is $-0.1\,\|a_t - a_{t-1}\|^2$, computed on the *raw* policy output (mjlab's `action_rate_l2`). Penalties are given as a share of the largest per-step tracking reward (the six tracking weights sum to 5; both are scaled by $\Delta t$).

**Data.**

- Figure 1: the `dev` references alone (1,294 clips), no policy.
- Figure 2: the two D27 A/B checkpoints (PPO-Sim, iteration 999, same seed), played the benchmark way on 96 stratified `dev` clips (first 10 s, deterministic, no noise; about 28k steps each).
- Only the left arm is shown; the right arm is the mirror. Two leg joints are included for scale.
- Scripts: `scripts/supermimic/action_stats.py`, `docs/supermimic/m1/make_action_scale_figures.py`.

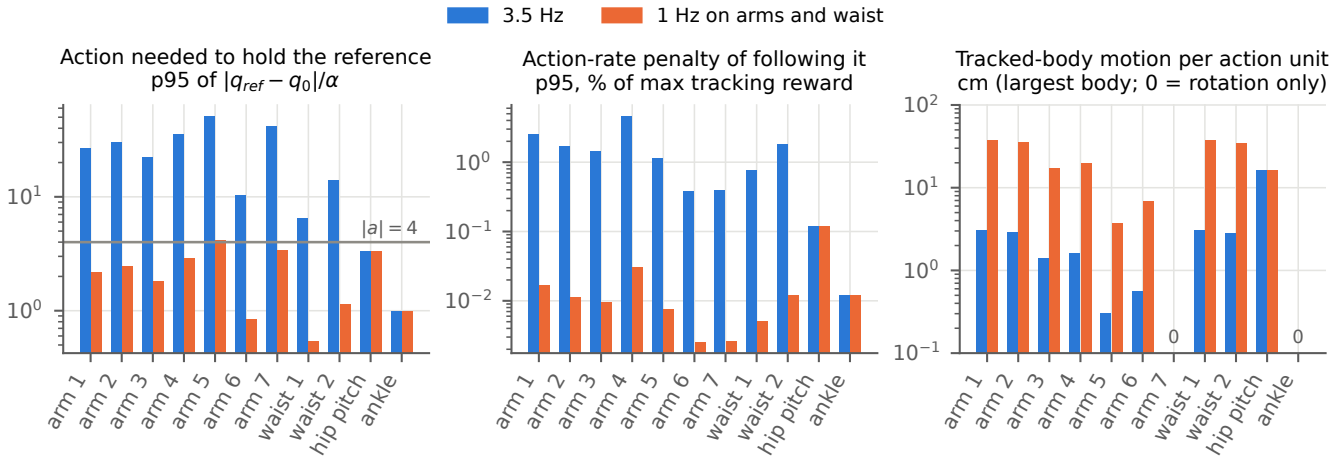



Figure 1: **What the references demand, per joint.** *Left:* action needed to hold the reference pose, $|q_{ref} - q_0|/\alpha$. At 3.5 Hz every arm joint needs 10–50 units (p95). *Middle:* action-rate penalty of following the reference. It is the same for all arm joints, ≈ 0.4–5% each at p95. *Right:* how far one action unit moves the tracked bodies, $\alpha\times$lever arm. Joints 1–4: 1.4–3.1 cm; joints 5, 6: 0.3, 0.6 cm; joint 7: 0, because it only rotates the wrist body about its own origin.

## Why the WBC report saw it mainly on joints 5–7

1. **The action-scale problem is arm-wide.** All seven arm joints have $\alpha \approx 0.04$–$0.06$ at 3.5 Hz. All need 10–50 units to hold their references, and all make following them expensive (Figure 1, left and middle). In the SuperMimic task the 3.5 Hz policy freezes the *whole* arm and the waist (Figure 2, bottom right).
2. **What differs per joint is how much reward one action unit can buy**, i.e. $\alpha\times$ lever arm $\times$ how much the reward weighs that body.
   - Joints 1–4 move the tracked elbow and wrist 1.4–3.1 cm per unit even at 3.5 Hz.
   - Joints 5 and 6 move them 0.3–0.6 cm per unit.
   - Joint 7 moves them 0 cm. It only rotates the wrist body, by $\alpha \approx 3.6°$ per unit, which is small against the orientation term's 0.4 rad (23°) scale averaged over the tracked bodies.
3. **The WBC task concentrates its arm signal on the end effector**: the target position is an actor input, and the end-effector body terms are rewarded. So the centimetre-scale effect of joints 1–4 is enough for PPO to learn them; joints 5–7, with almost no position effect and a flat orientation term, stay at $q_0$. SuperMimic tracks 14 bodies averaged together, of which the arm is 6, under the full action-rate penalty, so even joints 1–4 don't earn enough. This point is an interpretation consistent with both studies, not measured in the WBC task.

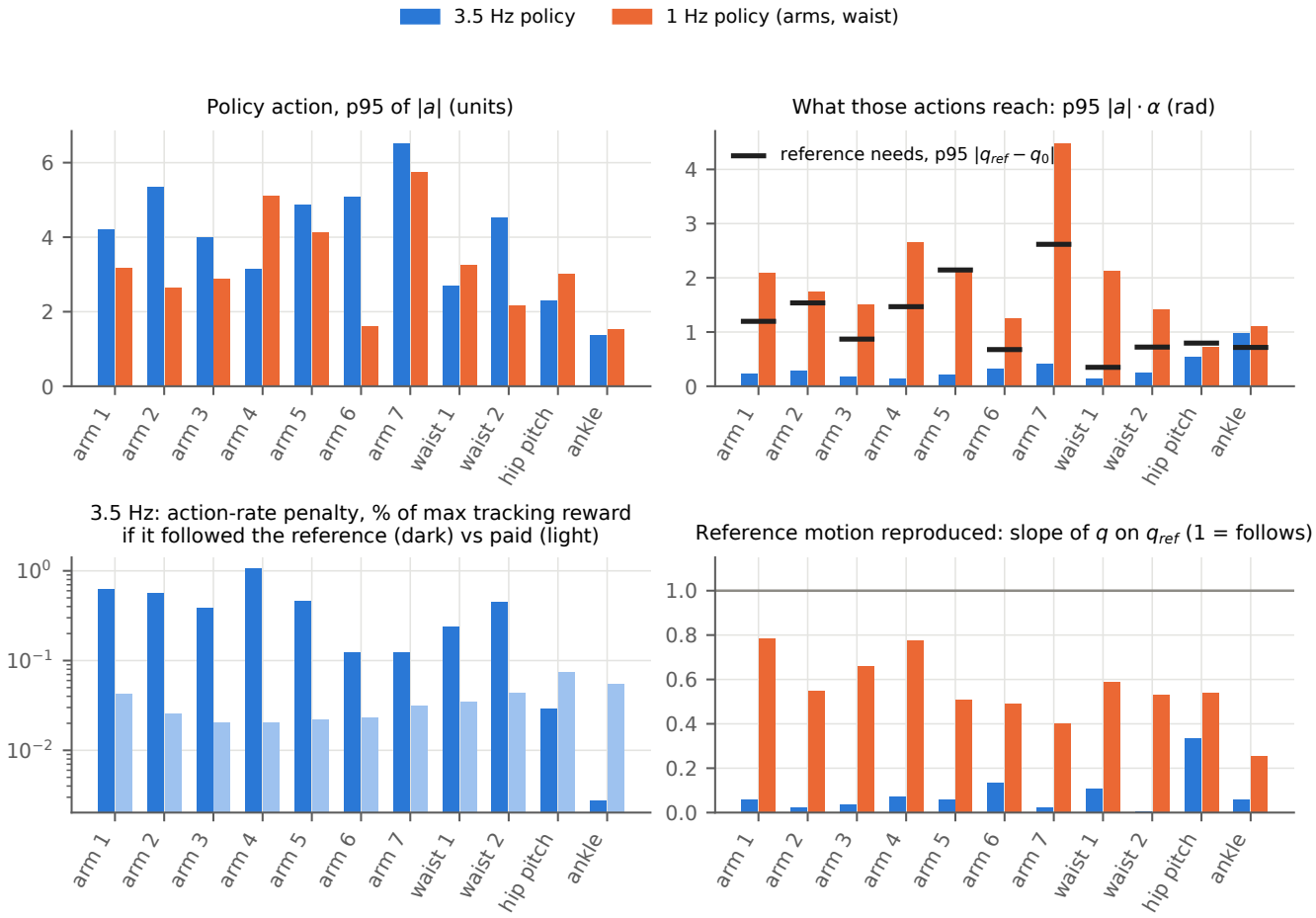



Figure 2: **What the trained policies do.** *Top left:* both policies keep $|a|$ within a few units (p95 3–6.5). *Top right:* at 3.5 Hz those units reach only 0.12–0.58 rad of setpoint, against the 0.68–2.62 rad the references need; at 1 Hz they reach 1.25–4.48 rad. *Bottom left:* at 3.5 Hz, following the references would cost 6.8% of the maximum tracking reward per step on average (p95 26%) for the arms alone. The policy pays 0.37%. *Bottom right:* share of each joint's reference motion the policy reproduces. At 3.5 Hz it is 0.01–0.14 for *every* arm and waist joint; at 1 Hz it is 0.40–0.78.

4. **The lever-arm effect stays at 1 Hz.** $\alpha$ grows 12× on every arm joint, but joint 7 still moves no tracked position. Its reproduction (0.40) is the lowest of the arm, and joint 5's error does not improve (0.89 vs 0.95 rad mean $|q - q_{ref}|$).

**Takeaway.** Choose $\alpha$ per joint from the reward the joint can affect, not only from $\tau_{\max}/K_p$. That is the WBC report's conclusion, and here it applies to the whole arm, with the wrist as the extreme case.